MGS 616_HW3_Group13

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1. Split the data into training (60%), and validation (40%) datasets. Run a multiple linear regression with the outcome variable Price and predictor variables Age_08_04, KM, Fuel_Type, HP, Automatic, Doors, Quarterly_ Tax, Mfr_Guarantee, Guarantee_Period, Airco, Automatic_airco, CD_Player, Powered_Windows, Sport_Model, and Tow_Bar.

Code:

```
# Select the columns for regression analysis
predictors = ['Age_08_04', 'KM', 'Fuel_Type', 'HP', 'Automatic', 'Doors', 'Quarterly_Tax',
              'Mfr_Guarantee', 'Guarantee_Period', 'Airco', 'Automatic_airco', 'CD_Player',
              'Powered_Windows', 'Sport_Model', 'Tow_Bar']
outcome = 'Price'
x = cars_df[predictors]
y = cars_df[outcome]

# Create dummy variables for the categorical column 'Fuel_Type'
x = pd.get_dummies(x, columns=['Fuel_Type'], drop_first=True)

# Split the data into training and validation sets (60% training, 40% validation)
x_train, x_valid, y_train, y_valid = train_test_split(x, y, test_size=0.4, random_state=42)

# Fit the regression model y on x
cars_lm = LinearRegression()
cars_lm.fit(x_train, y_train)

# Print the regression model coefficients
intercept = cars_lm.intercept_
coefficients = pd.DataFrame({'Predictor': x.columns, 'Coefficient': cars_lm.coef_})

# Display the intercept, coefficients
intercept, coefficients
```

Output:

```
(10139.352665041826,
           Predictor  Coefficient
 0          Age_08_04  -111.692000
 1                 KM    -0.017576
 2                 HP    29.732254
 3          Automatic   535.405917
 4              Doors   165.883438
 5      Quarterly_Tax    16.944377
 6      Mfr_Guarantee   156.521758
 7   Guarantee_Period    73.685120
 8              Airco   248.610630
 9    Automatic_airco  3015.495036
 10         CD_Player   304.590689
 11    Powered_Windows   384.759273
 12       Sport_Model   341.192746
 13           Tow_Bar  -283.741189
 14  Fuel_Type_Diesel  1785.071135
 15  Fuel_Type_Petrol  1997.005756)
```

a) **What appear to be the three or four most important car specifications for predicting the car's price?**

Based on the coefficients from the model, the three or four most essential car specifications for predicting the car's price are:

- **Automatic_airco (Coefficient: 3015.50)** - This has the highest positive impact on the car's pricing, showing that vehicles with automatic air conditioning are worth much more.
- **Fuel_Type_Petrol (Coefficient: 1997.01)** - When compared to other fuel kinds, petrol adds a significant value to the car's pricing.
- **Fuel_Type_Diesel (Coefficient: 1785.07)** - The diesel fuel type has a significant positive impact on the car's price.
- **Automatic (Coefficient: 535.41)** - A car with an automatic transmission costs a lot more money.

In linear regression, a **negative coefficient** means that as the predictor increases, the car's price decreases

- **Age_08_04 (Coefficient: -111.69):** As the car's age (months) increases, the price lowers. This is a common trend in the used car market, as older vehicles are typically less valuable owing to wear and tear, outdated features, and depreciation.

- **KM (Coefficient: -0.0176):** This shows that the price of the car drops as the mileage (kilometers driven) rises. A car with a higher mileage usually has more wear and tear on it, which lowers its resale value.

b) Using metrics you consider useful, assess the performance of the model in predicting prices.

Evaluation of Model Performance Using Various Metrics

After doing backward, forward, and stepwise feature selection, We used a subset of predictors to fit a multiple linear regression model.

A thorough evaluation of the model's ability to predict automobile prices using a variety of measures can be seen below.

Selected Predictors for the Model

The best subset of predictors, chosen through feature selection methods, includes:

Age_08_04, KM, Fuel_Type, HP, Automatic, Doors, Quarterly_ Tax, Mfr_Guarantee, Guarantee_Period, Airco, Automatic_airco, CD_Player, Powered_Windows, Sport_Model, and Tow_Bar.

Backward Elimination

- In this case, backward elimination did not remove any predictors, implying that all variables contribute significantly to the model.

Forward Selection:

- This method starts with no predictors and adds them one at a time, selecting the ones that improve the model the most. It eventually arrived at the same set of 16 predictors as backward elimination.

Stepwise Selection:

- A combination of both forward selection and backward elimination. It started with an empty model and added/removes predictors iteratively. The best subset identified was the same as that found in forward selection.

All three methods consistently identified the same 16 predictors as the most relevant for predicting car pricing, giving high confidence in the regression model's feature selection. This selection process guaranteed that the model contained only features that contributed significantly to explaining the variation in car pricing.

The model's performance was first assessed using the training dataset. Here are the key metrics obtained:

Training Set Performance:

- **Mean Error (ME): -0.0000** – The average residual (difference between actual and predicted prices) is extremely close to zero, indicating that the model's predictions are unbiased on the training data.
- **Root Mean Squared Error (RMSE): 1202.30** – This metric indicates that, on average, the predicted car prices deviate from the actual prices by around 1202 units. **Mean Absolute Error (MAE): 912.35** – The average absolute difference between the predicted and actual car prices is about 912 units.
- **Mean Percentage Error (MPE): -1.07%** – This value indicates that, on average, the model slightly underestimates car prices in the training set.
- **Mean Absolute Percentage Error (MAPE): 9.06%** – The model's predictions are off by approximately 9.06% on average relative to the actual car prices, indicating a reasonably accurate model.

Model Performance on Validation Data

To evaluate how well the model generalizes to new data, it was tested on the validation dataset. The key metrics for the validation set are as follows:

Validation Set Performance:

- **Mean Error (ME):** -116.82 – The model tends to slightly underestimate car prices on the validation set, with an average error of around -116.82 units. This suggests a small bias in the model.
- **Root Mean Squared Error (RMSE):** 1221.04 – On the validation set, the model's predictions have an average deviation of approximately 1221 units from the actual prices, which is close to the RMSE on the training set, indicating good model generalization.
- **Mean Absolute Error (MAE):** 890.80 – The average absolute difference between the predicted and actual prices in the validation set is around 891 units, similar to the training set's MAE.
- **Mean Percentage Error (MPE):** -2.07% – The model slightly underestimates car prices on the validation set by around 2% on average.
- **Mean Absolute Percentage Error (MAPE):** 8.76% – The model's predictions are off by approximately 8.76% relative to the actual prices on the validation set, indicating that the model maintains a good level of accuracy when predicting new data.

Screenshot of the output:

```
Intercept: 10139.352665041826
           Predictor  Coefficient
0          Age_08_04  -111.692000
1                 KM    -0.017576
2                 HP    29.732254
3          Automatic   535.405917
4              Doors   165.883438
5      Quarterly_Tax    16.944377
6      Mfr_Guarantee   156.521758
7   Guarantee_Period    73.685120
8              Airco   248.610630
9    Automatic_airco  3015.495036
10         CD_Player   304.590689
11    Powered_Windows   384.759273
12       Sport_Model   341.192746
13           Tow_Bar  -283.741189
14  Fuel_Type_Diesel  1785.071135
15  Fuel_Type_Petrol  1997.005756

Model performance on training data:

Regression statistics

                      Mean Error (ME) : -0.0000
       Root Mean Squared Error (RMSE) : 1202.2957
            Mean Absolute Error (MAE) : 912.3493
          Mean Percentage Error (MPE) : -1.0741
Mean Absolute Percentage Error (MAPE) : 9.0636

Model performance on validation data:

Regression statistics

                      Mean Error (ME) : -116.8193
       Root Mean Squared Error (RMSE) : 1221.0390
            Mean Absolute Error (MAE) : 890.7991
          Mean Percentage Error (MPE) : -2.0670
Mean Absolute Percentage Error (MAPE) : 8.7606
```